

Figure 1: Packet flow

Features

Listed below are some of the features of OVS:

- QOS
- Port mirroring
- NIC bonding
- VLAN
- GRE support

Installing OVS

To install on Ubuntu 12.04, give the following command:

```
apt-get install openvswitch-switch
```

After installing OVS, check the kernel module by using the following command:

```
lsmod|grep openvswitch
```

To view the version number, use the code below:

```
ovs-vsctl -V
ovs-vsctl (Open vSwitch) 1.4.0+build0
Compiled Feb 18 2013 13:13:22
```

Create the first bridge with the following command:

```
ovs-vsctl add-br bridge1
```

Add the virtual interface for virtual machines to the bridge:

```
ovs-vsctl add-port bridge1 vif1.0
ovs-vsctl add-port bridge1 vif2.0
```

...where vif1.0 is the virtual interface of vm1 with IP 10.168.122.3, and vif2.0 is the virtual interface of vm2 with IP 10.168.122.4

You can check the connectivity as follows:

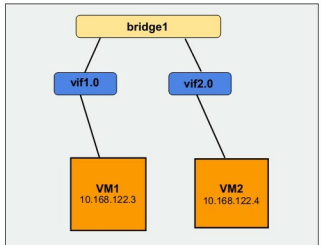


Figure 2: Block diagram

```
ping 10.168.122.4 -c 10
--- 10.168.122.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time
899ms
rtt min/avg/max/mdev = 0.140/0.243/0.615/0.130 ms
```

The corresponding data path flows are given below:

```
ovs-dpctl dump-flows bridge1
```

```
in_port(2),eth(src=52:9a:13:f4:90:c8,dst=ae:0a:5e:8a:cc:2b)
,eth_type(0x0800),ipv4(src=10.168.122.4,dst=10.168.122.3,pr
oto=1,tos=0,ttl=64,frag=no),icmp(type=0,code=0), packets:9,
bytes:882, used:0.690s, actions:1
```

```
in_port(1),eth(src=ae:0a:5e:8a:cc:2b,dst=52:9a:13:f4:90:c8)
,eth_type(0x0800),ipv4(src=10.168.122.3,dst=10.168.122.4,pr
oto=1,tos=0,ttl=64,frag=no),icmp(type=0,code=0), packets:9,
bytes:882, used:0.690s, actions:2
```

The above result is almost self-explanatory, where *in_port* can be seen from:

```
ovs-dpctl show

system@bridge1:
lookups: hit:42 missed:26 lost:0
flows: 0
port 0: bridge1 (internal)
port 1: vif1.0
port 2: vif2.0
```

Using VLAN for traffic isolation

Traffic between virtual machines can be isolated using VLAN. Openstack Neutron uses this feature of OVS to offer traffic isolation between different tenants by extending it over GRE tunnels.